

Abhay Kumar

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About Me

ML Professional with 12+ years building AI solutions in Generative AI, LLMs, Computer Vision, and NLP. Proven expertise architecting multi-tenant and autonomous systems, and driving measurable business impact.

Professional Experience

Sr AI Engineer

Target Corporation (Feb 2021 - Present)

Enterprise Agentic Document Intelligence Platform

- Led architecture and productionization of a multi-tenant, agentic document intelligence platform processing millions of documents/month across dozens of use cases, enabling template-agnostic extraction over heterogeneous document workflows.
- Designed a hybrid extraction + intelligent routing system (deterministic KV extraction + LLM fallback via query complexity classifier), reducing LLM usage ~30-50% vs LLM-only pipeline and achieving P95 latency of 6s, while sustaining 95+% extraction accuracy across semi-structured and unstructured documents.
- Architected orchestration, reasoning, tool discovery, and human-in-the-loop controls
- Established production standards: guardrails, tool gating, evaluation frameworks, monitoring, and observability for agent behavior.
- Built entity linking & validation layer integrated with internal enterprise systems, improving data consistency and reducing manual review effort, enabling downstream automation.

RAG Powered PDF Extractive QA Engine

- Engineered RAG pipeline with layout-aware semantic chunking, achieving 89% retrieval precision and improving answer relevance
- Architected the multi stage retrieval pipeline with Milvus, to query rightly chunked documents based on query intent and layout, and reranking to improve relevance

Multi-Modal Multi-Tenant Image Search Platform

- Designed and delivered an enterprise-grade multi tenant asset tagging and search platform supporting semantic and hybrid queries over millions of assets.
- Architected multi-modal hybrid search platform (text, image, keywords) using CLIP embedding with Milvus and Elasticsearch achieving 87% relevance across assets.
- Multi-tenant infrastructure with data isolation, RBAC, and tenant-specific configs;
- Onboarded 8+ internal teams across marketing, e-commerce, and content management
- Designed Offline ML-based metadata enrichment pipeline using OCR, object detection, logo recognition and leading to tag coverage increase across assets indexing.
- Implemented high-throughput batch and incremental asset tagging with Kafka, Spark, and distributed inference.
- indexed 1M+ assets with rich metadata through the pipeline

Senior Data Scientist

Embibe (May 2019 - Jan 2021)

Multimodal Question Tagging and Topic Extraction System

- Built automated systems for question tagging and topic extraction using BERT and ResNet Encoding model, improving content recommendation and search relevance
- Achieved 93% precision and 89% recall in question tagging, and 90% accuracy in topic extraction.

Senior Data Scientist

ClipIndia (Sep 2018 - Apr 2019)

Video Content Moderation & Metadata Enrichment Pipeline

- Deployed NSFW video moderation system reducing inappropriate content by 98%.
- Reduced manual moderation effort by 70% through ML-based metadata enrichment
- Built ML based video tagging models achieving 91% precision, 87% recall
- Fine-tuned 3DCNN models on 12 custom categories with ensemble approach combining fine-tuned + zero-shot pre-trained models for emerging categories

Senior Data Scientist

Cogknit Semantics (Sep 2015 - Sep 2018)

Catalogue Anomaly Detection Engine

- CV-based CNN system detecting out-of-category product placements, reducing catalogue errors by 92%;

Sr. Research Analyst

Edureka (Dec 2013 - Sep 2015)

- Created a comprehensive knowledge base, reducing TA team workload and improving support and learning resources.

Education

- Bachelor in Computer Science, Visvesvaraya Technological University (VTU), 2013.

Skills

- Generative AI: Prompt engineering, LLM app development, RAG, ChatGPT, LLAMA, CrewAI, LangChain, LangGraph, Langfuse , ADK
- AI Agents : Autonomous agents, multi-agent systems, Agent registry, Agent governance, agent orchestration, agent evaluation, agent monitoring, agentic workflow, human-in-the-loop controls, guardrails, tool gating
- Search Systems: Elasticsearch, Milvus, FAISS, CLIP, Sentence Transformers
- Computer Vision: Object detection, image classification, segmentation , LayoutLM, OpenCV, CNNs, 3DCNNs
- NLP: Text classification, topic modeling, intent classification, BERT, RoBERTa, Sentence Transformers, NLTK, SpaCy
- Machine Learning: Deep Learning (PyTorch, TensorFlow), Scikit-learn, Huggingface
- Applied ML: Model training, optimization, deployment at scale, algorithm research
- Data Processing: NumPy, Pandas, Spark, Hadoop
- Cloud Platforms: AWS, Azure, Google Cloud
- MLOps: Git, CI/CD, MLflow, Docker, APIs
- Team Skills: Team building, leadership, project planning, data storytelling

Publications and Presentations

- Poster: "Domain Adaptation of Image Caption Model for Video Descriptions" at AI & Deep Learning Conference (GTC 2018, NVIDIA).